

Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering
CSE 3102, Lab No. 10 , Lab Work

Consider the following instructions. Execute each one and write a report on the answers to the instructions. First complete the following tasks to create the database and the tables.

1. Create a database called “Hollywood_Movies”. Write the corresponding query for this in your report.

```
=> CREATE DATABASE Hollywood_Movies;
```

2. Create a table titled “Movies” with the following constraints.

```
=> CREATE TABLE Movies (  
    movie_id INT PRIMARY KEY,  
    title VARCHAR(100) NOT NULL,  
    release_year INT,  
    genre VARCHAR(50),  
    director_id INT,  
    box_office DECIMAL(12,2),  
    rating DECIMAL(3,1)  
);
```

3. Create a table titled “Directors” with the following constraints.

```
=> CREATE TABLE Directors (  
    director_id INT PRIMARY KEY,  
    director_name VARCHAR(100) NOT NULL,  
    birth_year INT,  
    awards_count INT  
);
```

4. Create a table titled “Actors” with the following constraints.

```
=> CREATE TABLE Actors (  
    actor_id INT PRIMARY KEY,  
    actor_name VARCHAR(100) NOT NULL,  
    birth_year INT,  
    awards_count INT  
);
```

5. Create a table titled “Movie_Actors” with the following constraints.

```
=> CREATE TABLE Movie_Actors (  
    movie_id INT,
```

```

    actor_id INT,
    role VARCHAR(100),
    is_lead_role BOOLEAN,
    PRIMARY KEY (movie_id, actor_id),
    FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),
    FOREIGN KEY (actor_id) REFERENCES Actors(actor_id)
);

```

6. Populate the table “Directors” with the following data.

```

=> INSERT INTO Directors VALUES
(1, 'Christopher Nolan', 1970, 15),
(2, 'James Cameron', 1954, 10),
(3, 'Steven Spielberg', 1946, 20),
(4, 'Greta Gerwig', 1983, 5),
(5, 'Quentin Tarantino', 1963, 12);

```

7. Populate the table “Actors” with the following data.

```

=> INSERT INTO Actors VALUES
(1, 'Leonardo DiCaprio', 1974, 10),
(2, 'Kate Winslet', 1975, 6),
(3, 'Cillian Murphy', 1976, 4),
(4, 'Margot Robbie', 1990, 3),
(5, 'Brad Pitt', 1963, 9),
(6, 'Tom Hanks', 1956, 12),
(7, 'Natalie Portman', 1981, 5),
(8, 'Christian Bale', 1974, 7),
(9, 'Matt Damon', 1970, 6),
(10, 'Emily Blunt', 1983, 4);

```

8. Populate the table “Movies” with the following data.

```

=> INSERT INTO Movies VALUES
(1, 'Inception', 2010, 'Sci-Fi', 1, 829.89, 8.8),
(2, 'Titanic', 1997, 'Romance', 2, 2187.46, 7.8),
(3, 'Oppenheimer', 2023, 'Biography', 1, 976.00, 8.7),
(4, 'Barbie', 2023, 'Fantasy', 4, 1445.69, 7.0),

```

```
(5, 'Pulp Fiction', 1994, 'Crime', 5, 213.90, 8.9),
(6, 'The Post', 2017, 'Drama', 3, 179.76, 7.2),
(7, 'Interstellar', 2014, 'Sci-Fi', 1, 696.74, 8.6),
(8, 'Dunkirk', 2017, 'War', 1, 526.94, 7.9),
(9, 'Avatar', 2009, 'Sci-Fi', 2, 2847.25, 7.8),
(10, 'Catch Me If You Can', 2002, 'Crime', 3, 352.11, 8.1);
```

9. Populate the table “Movie_Actors” with the following data.

```
=> INSERT INTO Movie_Actors VALUES
(1, 1, 'Dom Cobb', TRUE),
(2, 1, 'Jack Dawson', TRUE),
(2, 2, 'Rose DeWitt Bukater', TRUE),
(3, 3, 'J. Robert Oppenheimer', TRUE),
(3, 4, 'Jean Tatlock', FALSE),
(4, 4, 'Barbie', TRUE),
(5, 5, 'Vincent Vega', TRUE),
(6, 6, 'Ben Bradlee', TRUE),
(7, 8, 'Cooper', TRUE),
(8, 3, 'Tommy', TRUE),
(9, 2, 'Neytiri', FALSE),
(10, 6, 'Carl Hanratty', TRUE),
(10, 1, 'Frank Abagnale Jr.', TRUE);
```

Queries with solutions:

1. List all Sci-Fi movies and their ratings

```
=> SELECT title, rating
FROM Movies
WHERE genre = 'Sci-Fi';
```

				title	rating
☐				Inception	8.8
☐				Interstellar	8.6
☐				Avatar	7.8

2. Find all directors with more than 10 awards

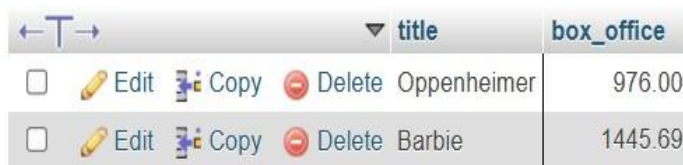
```
=> SELECT director_name, awards_count  
FROM Directors  
WHERE awards_count > 10;
```



	director_name	awards_count
☐ Edit Copy Delete	Christopher Nolan	15
☐ Edit Copy Delete	Steven Spielberg	20
☐ Edit Copy Delete	Quentin Tarantino	12

3. List movies released after 2010 and earned over 800 crore

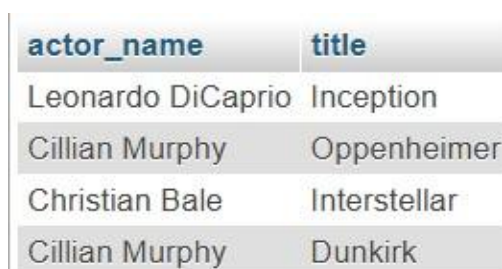
```
=> SELECT title, box_office  
FROM Movies  
WHERE release_year > 2010 AND box_office > 800;
```



	title	box_office
☐ Edit Copy Delete	Oppenheimer	976.00
☐ Edit Copy Delete	Barbie	1445.69

4. Show actors who played lead roles in Christopher Nolan's films

```
=> SELECT a.actor_name, m.title  
FROM Actors a  
JOIN Movie_Actors ma ON a.actor_id = ma.actor_id  
JOIN Movies m ON ma.movie_id = m.movie_id  
JOIN Directors d ON m.director_id = d.director_id  
WHERE d.director_name = 'Christopher Nolan' AND ma.is_lead_role =  
TRUE;
```



actor_name	title
Leonardo DiCaprio	Inception
Cillian Murphy	Oppenheimer
Christian Bale	Interstellar
Cillian Murphy	Dunkirk

5. Find all movies with a rating higher than 8.5

```
=> SELECT title, rating
FROM Movies
WHERE rating > 8.5;
```

		title	rating
☐	Edit	Inception	8.8
☐	Edit	Oppenheimer	8.7
☐	Edit	Pulp Fiction	8.9
☐	Edit	Interstellar	8.6

6. Show each director and the number of genres they have worked on

```
=> SELECT d.director_name, COUNT(DISTINCT m.genre) AS genre_count
FROM Directors d
JOIN Movies m ON d.director_id = m.director_id
GROUP BY d.director_id;
```

director_name	genre_count
Christopher Nolan	3
James Cameron	2
Steven Spielberg	2
Greta Gerwig	1
Quentin Tarantino	1

7. List actors who worked in more than 1 movie

```
=> SELECT a.actor_name, COUNT(ma.movie_id) AS movie_count
FROM Actors a
JOIN Movie_Actors ma ON a.actor_id = ma.actor_id
GROUP BY a.actor_id
HAVING COUNT(ma.movie_id) > 1;
```

actor_name	movie_count
Leonardo DiCaprio	3
Kate Winslet	2
Cillian Murphy	2
Margot Robbie	2
Tom Hanks	2

8: List movies with more than one lead actor

```
=>SELECT m.title, COUNT(*) AS lead_actor_count
FROM Movies m
JOIN Movie_Actors ma ON m.movie_id = ma.movie_id
WHERE ma.is_lead_role = TRUE
GROUP BY m.title
HAVING COUNT(*) > 1;
```

title	lead_actor_count
Catch Me If You Can	2
Titanic	2

9: Show actors who have acted in both Sci-Fi and Crime genres, along with the movie titles and genres

```
=>SELECT DISTINCT a.actor_name, m.title AS movie_title, m.genre
FROM Actors a
JOIN Movie_Actors ma ON a.actor_id = ma.actor_id
JOIN Movies m ON ma.movie_id = m.movie_id
WHERE a.actor_id IN (
    SELECT actor_id
    FROM Movie_Actors ma1
    JOIN Movies m1 ON ma1.movie_id = m1.movie_id
    WHERE m1.genre = 'Sci-Fi'
)
AND a.actor_id IN (
    SELECT actor_id
    FROM Movie_Actors ma2
    JOIN Movies m2 ON ma2.movie_id = m2.movie_id
    WHERE m2.genre = 'Crime'
)
AND m.genre IN ('Sci-Fi', 'Crime')
ORDER BY a.actor_name, m.genre;
```

actor_name	movie_title	genre
Leonardo DiCaprio	Catch Me If You Can	Crime
Leonardo DiCaprio	Inception	Sci-Fi

10. Show movie-wise total lead actors count

```
=> SELECT m.title, COUNT(*) AS lead_actors
FROM Movies m
JOIN Movie_Actors ma ON m.movie_id = ma.movie_id
WHERE ma.is_lead_role = TRUE
GROUP BY m.title;
```

title	lead_actors
Barbie	1
Catch Me If You Can	2
Dunkirk	1
Inception	1
Interstellar	1
Oppenheimer	1
Pulp Fiction	1
The Post	1
Titanic	2